

Sample Description of a Numerical variable

MATH 213

Key Features of the Distribution of a Numerical Variable

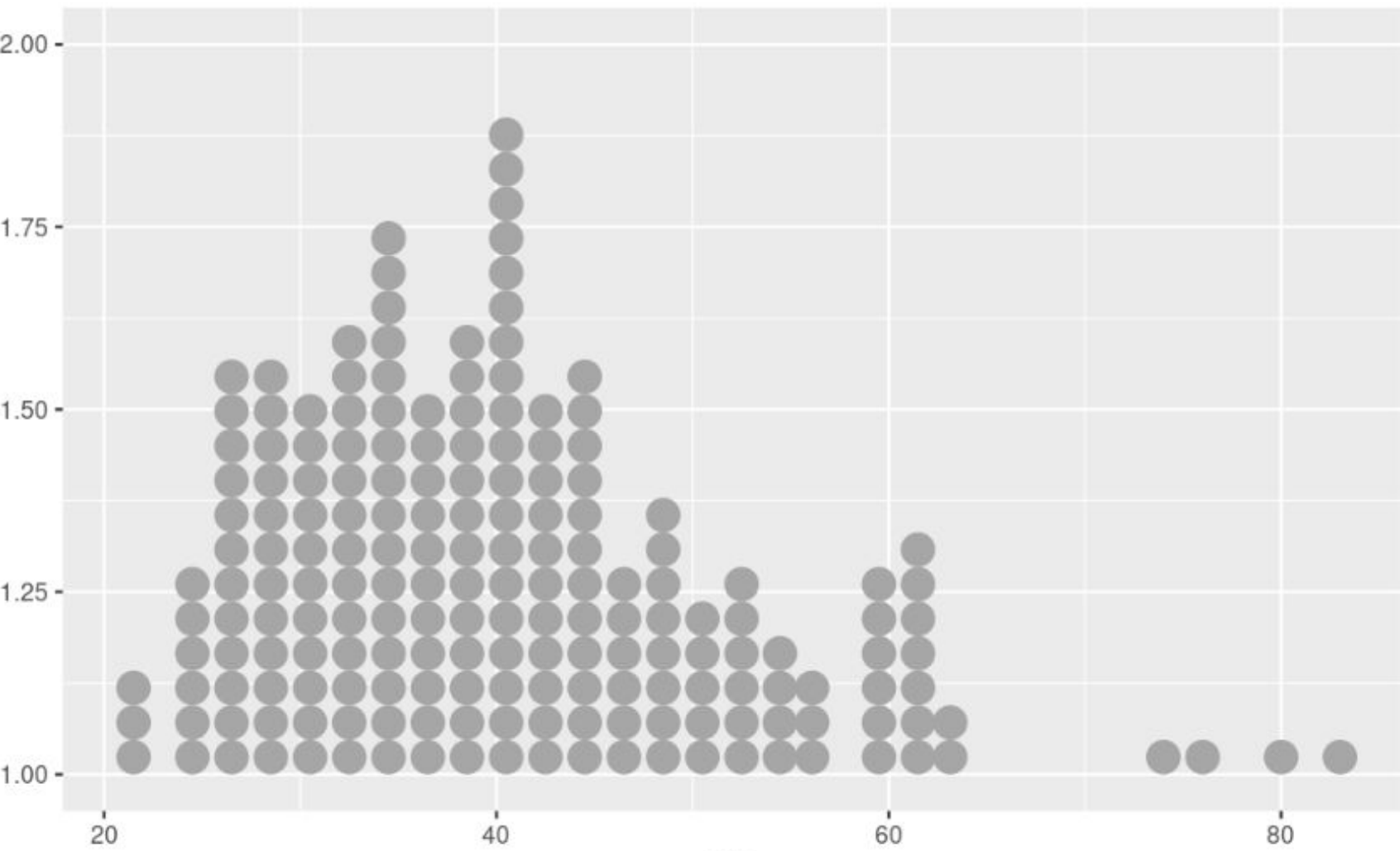
Every time we write a description of a numerical variable, we will address the following features:

- Who/what/where/when
- Shape
- Center
- Spread
- Outliers, if any
- Other interesting features, if any

Example

In class, we studied the distribution of the ages of Best Actor and Best Actress Oscar-winners. See a picture of the distribution on the next slide and see our sample description on the slide after that.

Notice: throughout the description, I am **using complete sentences** and I am speaking **in the context of Oscar-winners and their ages**. (I am not talking just about dots on the dotplot!) **If I use a technical term, such as “right-skewed”, I am also including language about what that means, in context.**



(Who/what/where/when) This graph displays the ages of all the Academy Awards best actor and best actress winners between 1929 and 2022.

(spread and center) The typical winner is about 40 years old, and the ages of winners range from the early twenties to over 80. Almost all winners are between about 25 and 65.

(shape) The winners ages are right skewed, meaning that the older winners are more spread out across a wider range of ages than the younger winners.

(outliers) We have four outliers that are significantly older than the rest.

Note: Above, I have included notes in purple about what features I am describing. When you are writing your own descriptions, you can leave these purple descriptors out and just write in paragraph form.

Interlude...

Let's now learn how to use R to calculate some summary statistics to help us with our descriptions...

Updated Sample Description

Using our calculations from R to update numbers below.

This graph displays the ages of all the Academy Awards best actor and best actress winners between 1929 and 2022. The typical winner is about 40 years old with a standard deviation of about 12. The ages of winners range from 21 to 83. The middle 50% of winners were between 32 and 47 years old, and almost all winners are between about 24 and 63. The winners ages are right skewed, meaning that the older winners are more spread out across a wider range of ages than the younger winners. Four winners are much older than the rest, including Anthony Hopkins at 83 and Jessica Tandy at 80.

Note: Above, I have included notes in purple about what features I am describing. When you are writing your own descriptions, you can leave these purple descriptors out and just write in paragraph form.